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1.  $A = 1011010$ ,  $B = 0101111$

$$\begin{array}{r} 1111 \\ 1011010 \\ + 0101111 \\ \hline 010001001 \end{array}$$

$$\begin{array}{r} 1011010 \\ - 0101111 \\ \hline 0101011 \end{array}$$

2.  $A+B$

$$\begin{array}{r} 2 \text{ C F 3} \\ + 0 \text{ 0 2 B} \\ \hline 2 \text{ D 1 E} \end{array}$$

$A-B$

$$\begin{array}{r} 2 \text{ C F 3} \\ - 0 \text{ 0 2 B} \\ \hline 2 \text{ C C 8} \end{array}$$

3.  $(174.25)_{10}$

$$\begin{array}{r|l} 2 & 174 \\ & 87 \\ & 43 \\ & 21 \\ & 10 \\ & 5 \\ & 2 \end{array} \begin{array}{l} 0 \\ 1 \\ 1 \\ 1 \\ 0 \\ 1 \\ 0 \end{array}$$

$$\begin{array}{r} 0.25 \times 2 = 0.5 \\ 0.5 \times 2 = 1 \end{array} \begin{array}{l} 0 \\ 1 \end{array}$$

$$\therefore (174.25)_{10} = 1010110.01$$

$$= AE.H$$

$$(5, 2, 5, 2, 0) \text{ MIP} = 209$$

$$(5+2+5)(5+1+2)(5+2+2)(5+2+2)(5+2+2) =$$

4.  $01101011$

$$\begin{array}{r} 10010100 \\ + 1 \\ \hline 10010101 \end{array}$$

5. A B C, S C<sub>0</sub> A ⊕ B ⊕ C

|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 |
| 0 | 1 | 0 | 1 | 1 |
| 0 | 1 | 1 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

Hence, from the truth table we can see that the function is same

6.  $f(a, b, c) = \sum m(1, 4, 5)$

| a | b | c | f |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 0 |

$$SOP = a'b'c + ab'c' + abc$$

(25.477)

|   |   |   |
|---|---|---|
| 0 | 0 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |
| 1 | 1 | 1 |
| 0 | 0 | 0 |
| 1 | 0 | 0 |
| 0 | 1 | 0 |
| 0 | 1 | 1 |

7.  $SOP = a'b'c + ab'c' + abc$

$POS = \prod M(0, 2, 3, 6, 7)$

$$= (a + b + c)(a + b' + c)(a + b' + c')(a' + b' + c)(a' + b' + c')$$

8.  $f(x_1, x_2, x_3) = x_1x_3' + x_2x_3' + x_1x_2x_3$

$$= x_1x_3'(x_2 + x_2') + x_2x_3'(x_1 + x_1') + x_1x_2x_3$$

$$= x_1x_3'x_2 + x_1x_3'x_2' + x_2x_3'x_1 + x_2x_3'x_1' + x_1x_2x_3$$

9.  $f(w, x, q) = (q + w')(x + q')(w + x + q)(w' + x')$

$$= (q + w' + xx')(x + q' + ww')(w' + x' + qq')$$

$$= (q + w' + x)(q + w' + x')(x + q' + w)(x + q' + w')(w' + x' + q)(w' + x' + q')(w + x + q)$$

10.  $f(a, b, c, d) = \sum m(0, 1, 4, 6, 9, 13, 14, 15)$

| ab \ cd | 00 | 01 | 11 | 10 |
|---------|----|----|----|----|
| 00      | 1  | 1  | 0  | 0  |
| 01      | 1  | 0  | 0  | 1  |
| 11      | 0  | 1  | 1  | 1  |
| 10      | 0  | 1  | 0  | 0  |

$$f = a'b'c' + a'bd' + abc + ac'd$$

11.  $f(a, b, c, d) = \sum m(1, 3, 6, 7) + d(4, 9, 11)$

| ab \ cd | 00 | 01 | 11 | 10 |
|---------|----|----|----|----|
| 00      | 0  | 1  | 1  | 0  |
| 01      | X  | 0  | 1  | 1  |
| 11      | 0  | 0  | 0  | 0  |
| 10      | 0  | X  | X  | 0  |

$$f = b'd + a'bc$$

12. List 1:  $(x+y+z)$  List 2:  $(x+y+z)$

|    |      |   |         |      |      |
|----|------|---|---------|------|------|
| 0  | 0000 | ✓ | (0,1)   | 000  | PI-1 |
| 1  | 0001 | ✓ | (0,4)   | 0-00 | PI 2 |
| 4  | 0100 | ✓ | (1,4)   | -001 | PI 3 |
| 6  | 0110 | ✓ | (4,6)   | 01-0 | PI 4 |
| 9  | 1001 | ✓ | (6,14)  | -110 | PI 5 |
| 13 | 1101 | ✓ | (9,13)  | 1-01 | PI 6 |
| 14 | 1110 | ✓ | (13,15) | 11-1 | PI 7 |
| 15 | 1111 | ✓ | (14,15) | 111- | PI 8 |

|      | 0 | 1 | 4 | 6 | 9 | 13 | 14 | 15 |
|------|---|---|---|---|---|----|----|----|
| PI 1 | x | x |   |   |   |    |    |    |
| PI 2 | x |   | x |   |   |    |    |    |
| PI 3 |   | x |   |   | x |    |    |    |
| PI 4 |   |   | x | x |   |    |    |    |
| PI 5 |   |   |   | x |   |    | x  |    |
| PI 6 |   |   |   |   | x | x  |    |    |
| PI 7 |   |   |   |   |   | x  |    |    |
| PI 8 |   |   |   |   |   |    | x  | x  |

EPI: PI 1, PI 4, PI 6, PI 8

$$f = a'b'c' + a'bd' + ac'd + abc \quad (\text{same as 10})$$

13.

List 1

|    |      |   |
|----|------|---|
| 1  | 0001 | ✓ |
| 4  | 0100 | ✓ |
| 3  | 0011 | ✓ |
| 6  | 0110 | ✓ |
| 9  | 1001 | ✓ |
| 7  | 0111 | ✓ |
| 11 | 1011 | ✓ |

List 2

|        |       |      |
|--------|-------|------|
| (1,3)  | 00-1  | ✓    |
| (1,9)  | -001  | ✓    |
| (4,6)  | 01-0  | PI 2 |
| (3,7)  | 0-11  | PI 3 |
| (3,11) | -011  | ✓    |
| (6,7)  | 011-1 | PI 4 |
| (9,11) | 10-11 | ✓    |

List 3

(1,3,9,11) | -0-1 | PI 1

|      | 1 | 3 | 6 | 7 |
|------|---|---|---|---|
| PI 1 | x | x |   |   |
| PI 2 |   |   | x |   |
| PI 3 |   | x |   | x |
| PI 4 |   |   | x | x |

EPI: PI 1, PI 4

$$\therefore f = b'd + a'bc \quad (\text{same as 11})$$

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| A \ BC | BC |    |    |    |
|--------|----|----|----|----|
|        | 00 | 01 | 11 | 10 |
| 0      | 0  | 1  | 0  | 0  |
| 1      | 0  | 1  | 1  | 1  |

$$f(A, B, C) = (B+C)(A+B')$$

$$= \Pi M(0, 2, 3, 4)$$

$$= (A+B+C)(A+B'+C)(A+B'+C')(A'+B+C)$$

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| A \ BC | BC |    |    |    |
|--------|----|----|----|----|
|        | 00 | 01 | 11 | 10 |
| 0      | 0  | 1  | 0  | 1  |
| 1      | 0  | 1  | 0  | 1  |

$$f(A, B, C) = \Pi M(0, 3, 4, 7)$$

$$= (B+C)(B'+C')$$